「Def.

Let $V$ be a real or Complex vector space.
A function $f: V\times V\to F$ is called a sesqui-linear form if it satisfies that
for all $\alpha,\beta,\gamma\in V$ and $c\in F$, $\begin{cases} f(c\alpha+\beta,\gamma)=c\cdot f(\alpha,\gamma)+f(\beta,\gamma) \\ f(\alpha,c\beta+\gamma)=\overline{c}\cdot f(\alpha,\beta)+f(\alpha,\gamma)\end{cases}$

In this page, we call the $f$ as form, for

「Thm.

Let $V$ be a finite dimensional inner product space and $f: V\times V\to F$ be a form.
Then, there is a unique linear operator $T$ on $V$ such that

$$f(\alpha,\beta)=\langle T\alpha,\beta\rangle \text{ for all } \alpha,\beta\in V.$$

Moreover, the map $f\mapsto T$ is isomorphism from space of forms onto $\mathcal{L}(V,V)$.

Proof. Let $\beta\in V$ be fixed. Then, a map $\alpha\mapsto f(\alpha,\beta)$ is a linear map on $V$.
By the Riesz-Representation theorem, for an orthonormal basis $\mathcal{B}=\{\alpha_1,\cdots,\alpha_n\}$,
there exists a unique vector $\beta'=\overline{f(\alpha_1,\beta)}\cdot\alpha_1+\cdots+\overline{f(\alpha_n,\beta)}\cdot\alpha_n$ such that

$$\langle\alpha,\beta'\rangle=f(\alpha,\beta) \text{ for all } \alpha\in V.$$

Define $U: V\to V;\beta\mapsto\beta'$. Then, given $\alpha,\beta,\gamma\in V$ and $c\in F$,

$$\langle\alpha,U(c\cdot\beta+\gamma)\rangle=f(\alpha,c\beta+\gamma)=\overline{c}\cdot f(\alpha,\beta)+f(\alpha,\gamma)=\overline{c}\cdot\langle\alpha,U(\beta)\rangle+\langle\alpha,U(\gamma)\rangle=\langle\alpha,cU(\beta)+U(\gamma)\rangle$$

Since the $\alpha$ was arbitrary, it proves $U$ is linear.
Moreover, since $V$ is finite-dimensional, so the adjoint exists, Put $T=U^*$.
Consequently, $f(\alpha,\beta)=\langle\alpha,U(\beta)\rangle=\langle T\alpha,\beta\rangle$. 」Existence.

Corollary. Given forms $f, g: V\times V \to F$, define

$$\langle f, g\rangle \overset{\text{def}}{=} \mathrm{Tr}(T_f T_g^*) \overset{(*)}{=} \sum_{j,k} f(\alpha_k, \alpha_j)\cdot \overline{g(\alpha_k, \alpha_j)}$$

is an inner product on $V\times V$ such that $(*)$, for any orthonormal basis $\{\alpha_1, \cdots, \alpha_n\}$ for $V$.

Proof. Given orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$, let $A = [T_f]_\beta$ and $B = [T_g]_\beta$.

Then, by the orthonormality and the above Thm,

$$A_{jk} = \langle T_f(\alpha_k), \alpha_j\rangle = f(\alpha_k, \alpha_j) \text{ and } B_{jk} = \langle T_g(\alpha_k), \alpha_j\rangle = g(\alpha_k, \alpha_j).$$

Moreover, $[T_f \circ T_g^*]_\beta = [T_f]_\beta \cdot [T_g^*]_\beta = AB^*$, So $\mathrm{Tr}(AB^*) = \sum_{j,k} A_{jk}\cdot \overline{B_{jk}}$ gives $(*)$.

Def. Let $V$ be a fin. dim inner product space, with basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$.

For a given form $f: V\times V \to F$, the matrix with entries

$$A_{jk} = f(\alpha_k, \alpha_j)$$

is called the matrix of $f$ in the ordered basis $\beta$.

Observe that: for a form $f: V\times V \to F$ and orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$..

Then, $f\left(\sum_s x_s\cdot \alpha_s, \sum_r y_r \alpha_r\right) = \sum_{r,s} \overline{y_r} A_{rs} x_s$, or

$$f(\alpha, \beta) = Y^* A X \quad \text{where } X = [\alpha]_\beta,\ Y = [\beta]_\beta.$$

Thm. Let $V$ be a fin. dim Complex inn. Prod Space.

For any form on $V\times V$, there is an orthonormal basis for $V$,

in which the matrix of $f$ is upper-triangular.

Proof. Since $V$ is fin. dim. Complex. inn. Prod, space, for linear operator $T: V\to V$ s.t

$$f(\alpha, \beta) = \langle T(\alpha), \beta\rangle,$$

it has an orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$ for $V$ s.t $[T]_\beta$ is upper-triangular. So,

$$f(\alpha_k, \alpha_j) = \langle T(\alpha_k), \alpha_j\rangle = 0 \quad \text{if } j > k.$$

Def. A form $f$ on $V$ is called Hermitian if

$$f(\alpha,\beta)=\overline{f(\beta,\alpha)} \quad \text{for all } \alpha,\beta\in V.$$

Note: if there is a linear operator $T$ on $V$ such that $f(\alpha,\beta)=\langle T(\alpha),\beta\rangle$, then

$$\overline{f(\beta,\alpha)}=\langle \alpha, T(\beta)\rangle=\langle T^*(\alpha),\beta\rangle, \quad \text{so,}$$

$f$ is Hermitian if and only if $T$ is Hermitian.

If $T$ is Hermitian, then $\langle T(\alpha),\alpha\rangle$ is real for all $\alpha\in V$, so $f(\alpha,\alpha)$ is real.

Lemma. Let $V$ be a complex inner product space, and $f$ is a form on $V$ such that

$$f(\alpha,\alpha) \text{ is real, for all } \alpha\in V.$$

Then, $f$ is Hermitian.

Proof. Given $\alpha,\beta\in V$,

$$f(\alpha+\beta,\alpha+\beta)=f(\alpha,\alpha)+f(\beta,\beta)+f(\alpha,\beta)+f(\beta,\alpha)$$
$$f(\alpha+i\beta,\alpha+i\beta)=f(\alpha,\alpha)+f(i\beta,i\beta)-i\cdot f(\alpha,\beta)+if(\beta,\alpha)$$

By the assumption, $f(\alpha,\beta)+f(\beta,\alpha)$ and $-i\cdot f(\alpha,\beta)+i\cdot f(\beta,\alpha)$ are real. Thus,

$$f(\alpha,\beta)+f(\beta,\alpha)=\overline{f(\alpha,\beta)}+\overline{f(\beta,\alpha)} \quad \text{and} \quad -i\cdot f(\alpha,\beta)+i\cdot f(\beta,\alpha)=i\cdot\overline{f(\alpha,\beta)}-i\cdot\overline{f(\alpha,\beta)}$$

Multiplying $i$ to the right, then we obtain

$$f(\alpha,\beta)-f(\beta,\alpha)=-\overline{f(\alpha,\beta)}+\overline{f(\alpha,\beta)}$$

So, $2f(\alpha,\beta)=2\cdot\overline{f(\alpha,\beta)}$, or $f$ is Hermitian. $\boxtimes$

Combining the above note with this Lemma, if $V$ is fin. dim, then

$T$ is Hermitian if and only if $\langle T(\alpha),\alpha\rangle$ is real.

「Principal Axis Theorem.

「Thm. For every Hermitian form $f$ on a fin. dim inn. prod space $V$, there exists an orthonormal basis $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ such that

$f$ is represented by a diagonal matrix with real entries.

Proof. Since $V$ is fin. dim, there exists a linear operator $T$ on $V$ s.t

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle.$$

And, by the lemma above, $f$ is Hermitian iff $T$ is Hermitian, so there is an orthonormal basis for $V$ which is consisted by eigenvector of $T$. Denote $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$, and $c_i$ be the eigenvalue corresponding to $\alpha_i$. Then,

$$f(\alpha_k, \alpha_j) = \langle T(\alpha_k), \alpha_j \rangle = c_k \cdot \delta_{kj},$$

thus proved. ($c_k$ are real by $T$ is Hermitian). ☒

Note: Under the condition hold,

$$f\left(\sum x_j \alpha_j, \sum y_k \alpha_k\right) = \sum_i c_i x_i \overline{y_i}$$

「HK. 9.2. (324P)

「Def.

A sesqui-linear form $f$ is said to be non-degenerate if: for fixed $\alpha \in V$,

$$\text{if } f(\alpha,\beta)=0 \text{ for all } \beta \in V, \text{ then } \alpha=0.$$

「Prop.

Let $f: V\times V \to F$ be a form on an inner product space $V$.

$f$ is non-degenerate if and only if the associated linear $T_f$ is non-singular.

Proof. Suppose that $f$ is non-degenerate. To show $T_f$ is non-singular, ($\ker T_f = \{0\}$) assume that $T_f(\alpha)=0$. That is, for any $\beta \in V$,

$$f(\alpha,\beta)=\langle T_f(\alpha),\beta\rangle=\langle 0,\beta\rangle=0$$

So, $\alpha=0$ by definition. Conversely, if $T_f$ is non-singular, then for any non-zero $\alpha \in V$,

$$f(\alpha,\beta)=\langle T_f(\alpha),\beta\rangle\neq 0 \quad \text{if } \beta=T_f(\alpha)$$

which proves $f$ is non-degenerate. ☑

Prop.

Let $f: V \times V \to F$ be a non-degenerate form on a finite dimensional inn. prod space $V$.
Given linear functional $L$, there exists one and only one vector $\beta \in V$ such that
$$L(\alpha) = f(\alpha, \beta) \text{ for all } \alpha \in V.$$

Proof By the theorem above, $f$ has invertible associated linear map $T$ such that
$$f(\alpha, \gamma) = \langle T(\alpha), \gamma \rangle \text{ for all } \alpha, \gamma \in V.$$

Meanwhile, by the Riesz Representation theorem, for the given $L$,
there exists a unique vector $\beta' = \overline{L(\alpha_1)} \cdot \alpha_1 + \cdots + \overline{L(\alpha_n)} \alpha_n$, ($\{\alpha_1, \cdots, \alpha_n\}$ is orthonormal)
$$L(\alpha) = \langle \alpha, \beta' \rangle \text{ for all } \alpha \in V.$$

Now, let $\beta = (T^*)^{-1}(\beta')$, then
$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle = \langle T(\alpha), (T^*)^{-1}(\beta') \rangle = \langle \alpha, (T^*) \circ (T^*)^{-1}(\beta') \rangle = \langle \alpha, \beta' \rangle = L(\alpha).$$

And, the uniqueness is given by: $f(\alpha, \beta_1) = L(\alpha) = f(\alpha, \beta_2) \Rightarrow f(\alpha, \beta_1 - \beta_2)$ and
Since $f$ is non-degenerate, so $\beta_1 - \beta_2$ must be zero. (Let $\alpha = \beta_1 - \beta_2$) ∎

Prop.

Let $f: V \times V \to F$ be a non-degenerate form on a finite dimensional inner product space $V$.
Given a linear operator $S: V \to V$, there exists an adjoint $S'$ relative to $f$.
(That is, $f(S\alpha, \beta) = f(\alpha, S'\beta)$ for all $\alpha, \beta \in V$).

Proof